

Topic 03 — SELECT & Filtering

PostgreSQL Bootcamp • A4 Revision Notes

SELECT

Retrieves the columns/data required. Example: `SELECT name, salary FROM employees;`

SELECT *

Means all columns. It does not specifically mean all rows.

Aliases

Use AS to give an output column a clearer name.

Expressions

Calculations can be performed inside SELECT, e.g. `salary * 12 AS annual_salary`.

DISTINCT

Removes duplicate result combinations.

WHERE

Filters rows using a condition.

Comparison

`=, <>, !=, >, <, >=, <=`

Logical Operators

AND requires all conditions; OR requires at least one; NOT reverses a condition.

NULL

Use `IS NULL` or `IS NOT NULL`. Do not use `= NULL`.

IN / NOT IN / BETWEEN

IN checks a list; NOT IN excludes values; BETWEEN includes both boundary values.

LIKE / ILIKE

% matches any number of characters. PostgreSQL ILIKE performs case-insensitive pattern matching.

ORDER BY

Sorts results. ASC = ascending; DESC = descending.

LIMIT / OFFSET

LIMIT restricts returned rows; OFFSET skips rows, useful for pagination.

Mental Model

SELECT → What? | FROM → Where? | WHERE → Which rows? | ORDER BY → Arrangement? | LIMIT → How many? | OFFSET → How many skipped?

Common Errors

Confusing columns with rows • using `= NULL` • missing quotes around text • incorrect AND/OR logic • misunderstanding BETWEEN • incorrect LIKE patterns • wrong sort direction.

Interview Takeaways

Be able to explain SELECT *, WHERE, DISTINCT, NULL handling, LIKE/ILIKE, ORDER BY, LIMIT and OFFSET in terms of what happens to columns and rows.